Z. 2026 **Concurrency Theory** P. Hofman

Tutorial 2, Friday, 09/10/2026, by VRM

Today: *LTL, CTL, and CTL**

Note. Unless stated otherwise, transition systems are finite Kripke structures and every state has at least one successor. LTL formulas are interpreted universally over all paths starting in the initial state.

A. Reading and writing temporal formulas

Exercise 1. Let *req*, *ack*, *error*, *reset*, *crit* and *ready* be atomic propositions. Write formulas expressing the following properties. For each property, state whether your formula is in LTL, CTL, or requires CTL*.

(a) Every request is eventually acknowledged.

(b) An error never occurs.

(c) From every reachable state, it is possible to reach a reset state.

(d) Whenever the system enters a critical section, it eventually leaves it.

(e) There exists a computation on which *ready* holds infinitely often.

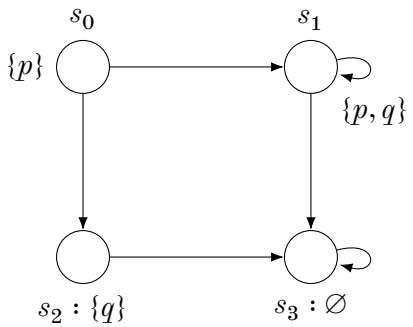
(f) Along every computation, from some point onward *ready* holds forever.

(g) From every reachable state there exists a computation on which *ready* eventually holds forever.

(h) Whenever *error* occurs, there exists a continuation on which *reset* is reached before the next occurrence of *error*.

B. Evaluate formulas on a transition system

Exercise 2. Consider the following Kripke structure.



For each formula below, determine the set of states satisfying it:

$$\mathsf{Sat}(\varphi) = \{s \mid s \models \varphi\}.$$

(a) $\mathrm{EX}\, q$

(b) $\mathrm{AX}\, q$

(c) $\mathrm{EF}\, q$

(d) $\mathrm{AF}\, q$

(e) $\mathrm{EG}\, p$

(f) $\mathrm{AG}\, p$

(g) $E(p\, U\, q)$

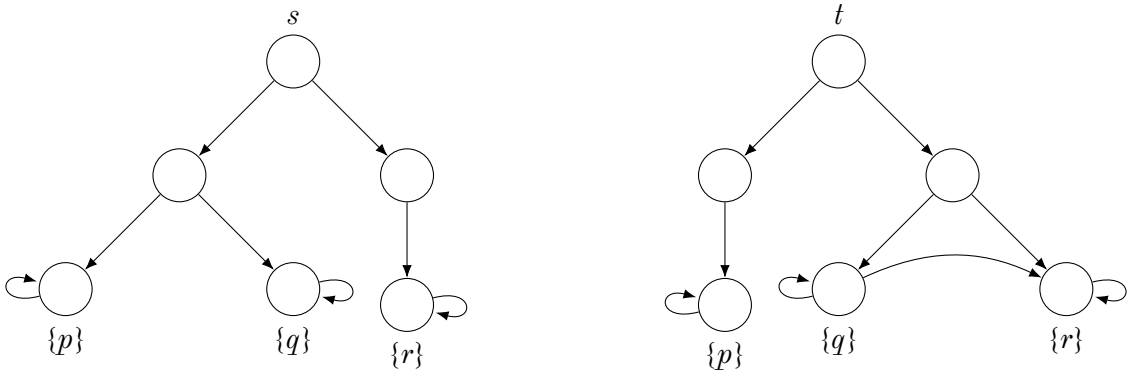
(h) $A(p\, U\, q)$

(i) $\mathrm{AG}\, \mathrm{EF}\, q$

For at least two formulas, compute $\mathsf{Sat}(\varphi)$ explicitly using the bottom-up CTL model-checking algorithm.

C. Find a distinguishing formula

Exercise 3. Consider the following two Kripke structures. All unlabeled states satisfy no atomic propositions.



(a) Find a CTL formula distinguishing s and t.

(b) Find such a formula of the smallest temporal depth you can.

(c) Show that the two systems do not have the same set of traces.

(d) Find an LTL formula distinguishing the two systems.

(e) Can an LTL formula distinguish the two systems when interpreted from s and t? Justify your answer in terms of traces.

Exercise 4. Construct two finite Kripke structures with the same set of traces from their initial states, but which are distinguished by some CTL formula.

Your answer should contain:

(a) the two transition systems;

(b) an argument that their trace sets are equal;

(c) a CTL formula distinguishing their initial states.

D. Equivalences and counterexamples

Exercise 5. Prove the following LTL equivalences or give a counterexample if they are not valid.

(a) $FF\varphi \equiv F\varphi$

(b) $GG\varphi \equiv G\varphi$,

(c) $FGF\varphi \equiv GF\varphi$,

(d) $GFG\varphi \equiv FG\varphi$,

(e) $X(\varphi \wedge \psi) \equiv X\varphi \wedge X\psi$,

(f) $F(\varphi \vee \psi) \equiv F\varphi \vee F\psi$,

(g) $G(\varphi \wedge \psi) \equiv G\varphi \wedge G\psi$.

For every proof, work directly from the semantics.

Exercise 6. For each proposed equivalence below, decide whether it is valid. If it is valid, prove it. Otherwise draw a counterexample.

(a) $F(\varphi \wedge \psi) \stackrel{?}{\equiv} F\varphi \wedge F\psi$

(b) $G(\varphi \vee \psi) \stackrel{?}{\equiv} G\varphi \vee G\psi$

(c) $\mathrm{EF}(\varphi \vee \psi) \stackrel{?}{\equiv} \mathrm{EF}\varphi \vee \mathrm{EF}\psi$

(d) $\mathrm{EF}(\varphi \wedge \psi) \stackrel{?}{\equiv} \mathrm{EF}\varphi \wedge \mathrm{EF}\psi$

(e) $\mathrm{AF}(\varphi \wedge \psi) \stackrel{?}{\equiv} \mathrm{AF}\varphi \wedge \mathrm{AF}\psi$

(f) $\mathrm{EG}(\varphi \wedge \psi) \stackrel{?}{\equiv} \mathrm{EG}\varphi \wedge \mathrm{EG}\psi$

(g) $\mathrm{AG}(\varphi \wedge \psi) \stackrel{?}{\equiv} \mathrm{AG}\varphi \wedge \mathrm{AG}\psi$

Exercise 7. Prove the following CTL dualities:

$$\neg\mathrm{EX}\,\varphi \equiv \mathrm{AX}\,\neg\varphi,$$

$$\neg\mathrm{EF}\,\varphi \equiv \mathrm{AG}\,\neg\varphi,$$

$$\neg\mathrm{AF}\,\varphi \equiv \mathrm{EG}\,\neg\varphi.$$

Which of the following are valid?

$$\mathrm{EX}\,\mathrm{EF}\,p \stackrel{?}{\equiv} \mathrm{EF}\,\mathrm{EX}\,p, \qquad \mathrm{AX}\,\mathrm{AF}\,p \stackrel{?}{\equiv} \mathrm{AF}\,\mathrm{AX}\,p.$$

Give a proof or a counterexample.

▷ More material here next time. ◁